1. Javascript & Uses

High level, interpreted Programming language that is primarily used to create dynamic and interactive content on the web. Javascript is a scripting language that runs in the browser (client-side) or on a server (with environment like node.js

Uses

1. Web development
2. Front-End & Back-End
3. Framework & Libraries
4. Real time Applications.
   Eg: Chat applications
5. Integration with API

2. Variable

Contained for storing data values

1. Var (Traditional Declaration
2. let (Block-scoped Declaration)
3. const (Constant declaration)

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3. Data Types

Categories that determine what kind of value a variable can hold.

1. Primitive
2. Reference

Eg: Number, string, Boolean, Object, Array

4. Diff. b/w var, let, const

var	let	const
Function scope	Block scope	Block scope
Hoisted (initialized to undefined)	Hoisted (Temporal Dead zone until declared	Hoisted (TDZ until declared)
Redeclaration Allowed in same scope	Not Allowed in same scope	Not Allowed in same scope
Reassignment Allowed	Allowed	Not-allowed

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5. Operators

Arithmetic : Perform mathematical operations

Comparison : Compare two values return a boolean

Logical operators : Combine multiple conditions to return a boolean result

6. == & ===

Loose equality : == - Performs type coercion meaning it converts the values to same type before making comparison

Compares values after trying to convert them to a common type

Strict equality === : operator compares both value and type No type coercion is done

It only returns true if both the value and type are same

7. Function : Block of Reusable code designed to perform a specific task

1. Code Reusability
2. Modularization
3. Maintainability

8.
```
function add(a,b) {
    return a+b;
}
console.log(add(5,3));
```

9. Array is a special type of object in Javascript used to store multiple values.

Two ways

1. Using Array Literals
2. Using Array Constructor

10. Accessing elements in an Array

We can access elements in an array using their index starting from zero

11. Object : Collection of Key-value pairs each key (also called a property) is a string (or symbol) and each value can be any valid datatype including another object or function.

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12. Accessing values in object

Use . dot notation to access the property of object

Use square brackets [ ] and pass the property name as a string

This is useful when property name is dynamic

13. Array of Objects.

Collection of Objects stored in an array. Each element of the array is an object and each object can have its own set of key-value pairs

14. for (let i=0; i<people.length; i++{
console.log(people[i].name);
}

15. Arrays : Collection of items where the order matters

Perform operations like sorting, filtering or looping through elements

Objects : Store data that is with unique keys

Need to represent structured data where each property has unique identifier

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